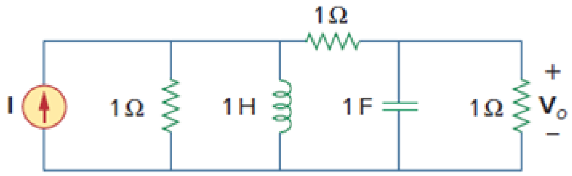
Chapter 14, Problem 78P

Problem

Redesign the circuit in Fig. 14.85 so that all resistive elements are scaled by a factor of 1,000 and all frequency-sensitive elements are frequency-scaled by a factor of 104.

Figure 14.85:



Step-by-step solution

Step 1 of 2

$$R' = K_m R = (1000)(1) = 1K\Omega$$

$$L' = \frac{K_m}{K_f} L = \frac{10^3}{10^4}(1) = 0.1H_Z$$

$$C' = \frac{C}{K_m K_f} = \frac{1}{(10^3)(10^4)} = 0.1\mu F$$

The new circuit is shown below:-

Step 2 of 2